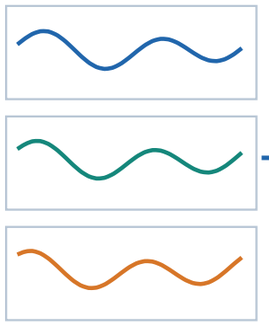
JOINTZIP | ENCODING FRAMEWORK

Source window x + user requirements: E (ε), R (ρ), A (τ, lags)

Time-series archive



Window by window

1 Window cap search

Window statistics
→ starting scale a(x)

λmin — **a(x)** — λmax = ε

Propose and refine λ ≤ ε
One scalar cap
per window and trial

x, λ

Compatible codec

REPLACEABLE BACKEND

Encode(x, λ) → B

Decode(B) → b

Verify requested cap λ

b, B

2 Joint correction



Decoded b → y

E: value error
R: correlation
A: lag correlation

Check E/R/A → save edits F

No edits if the base already passes

Measured base + correction bytes guide the next cap

Lossy + lossless candidates → smallest measured record → decode + E/R/A check

selected bytes

Choose one backend: SZ3 / ZFP / SPERR-1D / Serf-XOR / Serf-QT / SZ2016

Absolute-error control • reproducible source-free decoding • decoded-value access

Stored window record

H | base B | F

Receiver decoding

Decode B, apply F; no x

Decoded y



Reuse across analytical tasks